

135. Candy

Solved 

Hard

 Topics

 Companies

There are n children standing in a line. Each child is assigned a rating value given in the integer array `ratings`.

You are giving candies to these children subjected to the following requirements:

- Each child must have at least one candy.
- Children with a higher rating get more candies than their neighbors.

Return the minimum number of candies you need to have to distribute the candies to the children.

Example 1:

Input: `ratings = [1,0,2]`

Output: 5

Explanation: You can allocate to the first, second and third child with 2, 1, 2 candies respectively.

Example 2:

Input: `ratings = [1,2,2]`

Output: 4

Explanation: You can allocate to the first, second and third child with 1, 2, 1 candies respectively.

The third child gets 1 candy because it satisfies the above two conditions.

CANDY

Question: There are 'n' children standing in a line. Each child has a rating.

- Each child must get one candy atleast
- Children with higher rating must get higher candy than their neighbors



A

A GREEDY SOLUTION

5	4	3	5	6	2
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look_left_neighbors

↓	↓	↓	↓	↓	↓
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A

A GREEDY SOLUTION

5	4	3	5	6	2
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look_left_neighbors

1	1	1	2	3	1
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A

A GREEDY SOLUTION

-5

5	4	3	5	6	2
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3	2	1	2	3	1
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= (12) Avg

A

```
1  class Solution {
2  public:
3      int candy(vector<int>& ratings) {
4          int n=ratings.size();
5          vector<int> candies(n,1);
6
7          for (int i=1;i<n;i++){
8              if(ratings[i]>ratings[i-1]){
9                  candies[i]=candies[i-1]+1;
10             }
11         }
12
13         for (int i=n-2;i>=0;i--){
14             if(ratings[i]>ratings[i+1]){
15                 candies[i]=max(candies[i],candies[i+1]+1);
16             }
17         }
18         int result=0;
19         for (int x:candies){
20             result+=x;
21         }
22         return result;
23     }
24 };
```